RÉSUMÉ · CURRICULUM VITAE

James M. Rosado, *Ph.D.*

Applied Research Mathematician · Software Integration Engineer

Ten years of turning mathematics into working software: HPC infrastructure automation, AI model evaluation and machine learning at the National Security Agency, and numerical PDE solvers, neuron simulations and VR software in academic research. Former high-school and college mathematics teacher.

jarosado0911@gmail.com github.com/jarosado0911 Maryland, USA English · Spanish

TS/SCI clearance since 2021

2025	8	Ph.D.	7	1.8M	10+
NSA Director's Award for contributions to AI standards and research	DoD/NSA awards since 2023, for software and AI work	Mathematics, Temple University, 2022	publications: SIAM, PLOS Comp. Biol., Brain Stimulation, LAA	NSF supercomputing hours: XSEDE awards, three years running	years of research and software engineering

Awards & honors

NATIONAL SECURITY AGENCY · 2025

NSA Director's Award

For contributions to AI standards and research at the agency.

DoD / NSA

- **2025** Time Off Award: development of effective AI standards
- **2025** Monetary Award: software development of AI analysis tools
- **2025** Monetary Award: improved software performance at the agency
- **2024** Monetary Award: article recommendation software
- **2024** Performance Coin Award: ML-based recommendation systems
- **2023** Time Off Award: advanced coding practices
- **2023** Monetary Award: advanced query-based algorithms

National Science Foundation (XSEDE)

- **2022** Compute Research Award, 500,000 core-hours on EXPANSE ($2,440)
- **2021** Compute Research Award, 500,700 core-hours on EXPANSE ($2,440)
- **2020** Compute Research Award, 800,000 service units on COMET ($12,873)

Project DMS200031, "Ultrastructural Calcium Simulations in Neurons", San Diego Supercomputer Center.

Temple University

- **2022** Jay Novik Graduate Student Fellowship, for exceptional performance ($5,000)
- **2022** Dissertation Completion Grant ($8,750)
- **2020** First Summer Research Initiative Award ($6,000)

Rowan & Rutgers

- **2016** Certificate of Achievement in Mathematics: JMM participation and publishing
- **2015** Certificate of Achievement in Mathematics: JMM participation

2019	NIH BRAIN Initiative Summer School funding award
2019	Mathematics Department Excellence in Teaching Award ($500)
2018	SIAM recognition: leadership of SIAM Chapter activities
2003–07	Edward J. Bloustein Distinguished Scholar, Rutgers

Experience

Aug 2025 – present

Software Integration Engineer 4

SageCor Solutions · Annapolis Junction, MD

Automation team: setting up, deploying and optimizing HPC infrastructure as code.

- Designed and deployed Ansible playbooks and roles tailored to customer hardware; AWX job templates for HPC provisioning and node management.
- Built Ansible roles for lights-out management (LOM) of HPE and Dell servers, including user roles and accounts.
- Developed Python modules interfacing Ansible with power distribution units (PDUs) for user account management.
- Implemented and managed Keycloak identity and access management; improved credential management and user tracking.
- Implemented CI/CD pipelines to industry standards; code reviews in Git; Agile delivery tracked in Jira.
- Improved hardware device communication with NetBox by implementing and debugging Ansible code.
- Optimized the Ansible code base to speed hardware onboarding and the accessibility of HPE and Dell servers.
- Applied infrastructure-as-code principles for efficiency and scalability; implemented and tested on the Linux CLI.

Sep 2022 – Sep 2025

Applied Research Mathematician

National Security Agency · U.S. Department of Defense · Maryland

Research and software across several branches of the agency, with internal publications read across DoD divisions. TS/SCI clearance. Rotations:

Cybersecurity Data Scientist — Apr – Aug 2025

- Implemented a Python module with Elasticsearch for query searches of data.
- Designed and implemented a C++ wrapper class of Elasticsearch for streamlined access to its capabilities.
- Built and tested Python and C++ code with GitLab CI/CD.

Operations Data Scientist — Aug 2024 – Apr 2025

- Evaluated AI models and wrote Python code to deconstruct them; analyzed their strengths and weaknesses.
- Built matrix analysis and AI evaluation tools with TensorFlow, PyTorch and Plotly.
- Published an internal paper on the relationships between AI model layers and linear transformations.
- Monetary Award for AI analysis tools; Time Off Award for effective AI standards.

Operations Software Engineer — Jan – Aug 2024

- Implemented a code class for optimizing execution; compared the efficiency of coding structures.
- Found structures that use HPC memory and CPU resources efficiently; code released for IC use to expedite mission outcomes.
- Monetary Award for improved software performance.

Engagement and Policy

Sep 2023 – Jan 2024

- Built article recommendation systems with TF-IDF and word2vec neural networks in Python, speeding the onboarding of new hires.
- Promoted the software across agency organizations and produced usage data for agency offices.
- Presented at an internal agency conference to IC divisions; authored an internal technical paper.
- Performance Coin Award for ML-based recommendation systems.

Mathematics Research

Oct 2022 – May 2023

- Implemented graph-based query algorithms: nearest-neighbor descent and distributed hierarchical navigable small-world graphs.
- Worked with the developers of graph-based query software at Lawrence Livermore National Laboratory.
- Developed templated code for general use in query-based software, tested and validated on HPC systems.
- Time Off Award and Monetary Award for software engineering performance.

Summers 2023 & 2025

Summer Conference in Applied Mathematics Problems

Institute for Defense Analyses (IDA) · Princeton, NJ (2023) and Bowie, MD (2025)

- 2025: analyzed boolean logic circuits in FPGAs and implemented a Python boolean circuit analyzer to aid FPGA reverse engineering.
- 2023: built parallel software for processing large audio data, with statistical methods and bispectrum analysis for audio change-point detection; authored an internal article.
- Presented to technical audiences; organized code and tests in GitLab with CI/CD.

Summer 2021

Graduate Mathematics Program, summer intern

National Security Agency · Maryland

- Implemented and measured neural networks and binary classifiers that detect synthesized speech.
- Developed detection algorithms using bispectrum analysis and topological data analysis.
- Prepared a briefing on the results for the Director of the National Security Agency; co-authored an internal technical paper.
- Granted a Top Secret/SCI clearance.

Sep 2019 – Sep 2022

Researcher, Center for Computational Mathematics & Modeling (C2M2)

Temple University · Philadelphia, PA

Led development of Neuro-VISOR, real-time interactive simulation of neurons in virtual reality.

- Implemented sparse numerical solvers for the Hodgkin-Huxley PDEs of neuron electrical signaling, and synapse models for small networks.
- Parallelized the solvers with threads; C# and Unity3D VR.
- Used in neuroscience classrooms and in simulations of small neuronal networks.
- Wrote Python mesh generation for VR neuron meshes by parallel transport of contours.
- Led weekly team meetings and coordinated the project on GitHub.

Sep 2018 – Sep 2022

Researcher, Queisser Research Group

Temple University · Philadelphia, PA

- Co-developed software modeling neuron calcium dynamics under the electric fields of transcranial magnetic stimulation (TMS).

- Built ultrastructural spine-to-dendrite calcium simulation pipelines in ug4: finite volume/element methods, domain decomposition and MPI.
- Won NSF XSEDE compute three years running (1.8M hours on SDSC EXPANSE and COMET).
- Collaborated with the University of Minnesota and the University of Freiburg; mentored students and led project meetings.
- NIH funding to attend the BRAIN Initiative summer school at the University of Missouri.

2009 – 2022

Teaching

Temple University · Rowan University · Clearview Regional School District

- **Graduate Teaching Assistant**, Temple (2017–2022): taught and graded mathematics courses, designed syllabi and exams, ran Blackboard and Canvas sites, tutored calculus and algebra.
- **Adjunct Professor**, Rowan (2016–2017): evening Calculus I; lectures, notes, assessments and review sessions.
- **High School Mathematics Teacher**, Clearview Regional (2009–2017): grades 9–12; NJ core standards, co-taught with special education, IEP and 504 accommodations, mentored Rowan education majors.
- **Private Tutor**, Gloucester County, NJ (2011–2016).

Sep 2012 – Jun 2016

Mathematics Researcher

Rowan University · Glassboro, NJ

- Developed an algorithm to partition finite frames (linear algebra), implemented in Mathematica and published.
- Developed a method for evaluating non-trivial definite integrals with complex analysis.
- Presented at the Joint Mathematics Meetings and university symposia.

Skills

Languages

C/C++ · Python · Shell · C# · MATLAB · Mathematica · Java · Lua

HPC & infrastructure

Ansible · AWX · CI/CD · MPI · OpenMP · Keycloak · NetBox · Linux (RHEL, CentOS)

AI, ML & data

PyTorch · TensorFlow · scikit-learn · word2vec · Elasticsearch · Plotly

Numerical & scientific

Numerical PDEs · Finite volume / element · Sparse solvers · Domain decomposition · ug4 · Unity3D

Tools

Git · GitHub · GitLab · Bitbucket · Jira · LaTeX · Sage · Maple

Communication

Technical briefings · Teaching & curriculum · Mentoring · English & Spanish

Education

2017 – 2022
Ph.D., Mathematics
Temple University, Philadelphia, PA

Dissertation: "Ultrastructural Neuronal Modeling of Calcium Dynamics under Transcranial Magnetic Stimulation"

2022 – 2025
Cryptologic Mathematician Certificate
National Security Agency, Maryland

2012 – 2016
M.A., Mathematics
Rowan University, Glassboro, NJ

Thesis: "Partitions of Finite Frames"

2008 – 2009
Teacher of Mathematics Certificate
New Jersey Department of Education

2003 – 2007
B.S., Electrical & Computer Engineering
Rutgers University, New Brunswick, NJ

Publications

01 Neuronal resilience and calcium signaling pathways in the context of synapse loss and calcium leaks: a computational modeling study and implications for Alzheimer's disease

J. M. Rosado, P. R. Borole, M. Neal, G. Queisser. *SIAM Journal on Applied Mathematics* 83(6), 2418–2442, 2023.

02 Calcium modeling of spine apparatus-containing human dendritic spines demonstrates an “all-or-nothing” communication switch between the spine head and dendrite

J. M. Rosado, V. D. Bui, C. A. Haas, J. Beck, G. Queisser, A. Vlachos. *PLOS Computational Biology*, 2022.

03 Ultrastructural neuronal modeling of calcium dynamics under transcranial magnetic stimulation

J. M. Rosado. Doctoral dissertation, Temple University, 2022.

04 Multiscale modeling toolbox for single neuron and subcellular activity under transcranial magnetic stimulation

J. M. Rosado, S. Shirinpour, N. Hananeia, H. Tran, C. Galanis, A. Vlachos, P. Jedlicka, G. Queisser, A. Opitz. *Brain Stimulation* 14(6), 1470–1482, 2021.

05 Partitions of equiangular tight frames

J. M. Rosado, H. D. Nguyen, L. Cao. *Linear Algebra and its Applications* 526, 95–120, 2017.

06 **Partitions of finite frames**

J. M. Rosado. Master's thesis, Rowan University, 2016.

07 **A table of definite integrals from the marriage of power and Fourier series**

J. M. Rosado, T. Osler. *Scientia* 26, 77–82, 2015.

Plus internal technical papers at the NSA and IDA (AI model analysis, recommendation systems, audio change detection, synthesized speech detection).

Open-source software

2018–2022 · C# · UNITY

Neuro-VISOR

Real-time simulation of neuron electrical dynamics in VR (Temple C2M2).

2018–2019 · MULTISCALE

NeMo-TMS

Multiscale modeling of TMS effects on single neurons (Opitz Lab, Minnesota).

2021–2023 · MATLAB

CalcSim

Intracellular calcium dynamics of single neurons (Queisser Group).

2022–PRESENT · PYTHON

PythonNeuronMeshes

Surface meshes of neurons from their 1D models.

2026 · C++17

NeuronMesh

Closed neuron surfaces and Neuro-VISOR archives from reconstructions.

2026 · PYTHON · QT

SpineBuilder

Simulation meshes of ideal and reconstructed dendritic spines and whole cells.

Presentations

Jun 2022 Ultrastructural Neuronal Modeling of Calcium Dynamics Under Transcranial Magnetic Stimulation. Doctoral defense, Temple University.

Oct 2021 Neuron Dendritic Spines: Modeling Calcium Communication. Temple College of Science and Technology Research Mixer.

May 2021 Hodgkin-Huxley Conductance Based Model: From 1D to 3D. Temple Numerical PDEs course.

Jan 2020 An Investigation of Spine to Dendrite Calcium Communication. Temple Applied Mathematics Seminar.

Oct 2019 Inner Workings of a Neuron: A Mathematical Perspective. Temple Math Club.

May 2019 A Walkthrough Calculus of Variations. Temple Methods in Applied Mathematics course.

Dec 2018 Sturm-Liouville Theory: An Example. Temple Methods in Applied Mathematics course.

Mar 2018 A table of definite integrals from the marriage of power and Fourier series (with T. Osler). EPaDel Spring Section Meeting, Philadelphia.

Fall 2017 Partitions of Equiangular Tight Frames. Temple Graduate Student Seminar.

Jan 2021 Applied Mathematics: Modeling Neuronal Electrical and Ion Dynamics. Temple University.

Dec 2020 PDE Based Image Reconstruction. Temple Numerical PDEs course.

Nov 2020 Modeling an Action Potential and Neuronal Behavior. Temple Graduate Seminar.

Jan 2017 Partitions of Equiangular Tight Frames (with H. Nguyen, L. Cao). Joint Mathematics Meetings, Atlanta, GA.

May 2016 Partitions of Equiangular Tight Frames (with H. Nguyen, L. Cao). Rowan STEM Symposium.

Apr 2016 Frame Partitioning Algorithms. Master's thesis presentation, Rowan University.

Aug 2015 A table of definite integrals from the marriage of power and Fourier series (with T. Osler). MAA MathFest, Washington, D.C.

May 2015 The same work at the Rowan STEM Symposium and the Sigma Xi Research Symposium, St. Joseph's University.

Jan 2015 The same work at the Joint Mathematics Meetings, San Antonio, TX.

More

Undergraduate research

2005 · Rowan College of Engineering: transistor-based replica of a leech heart interneuron, from Hodgkin-Huxley formulas; FPGA software for teaching.

2004 · Universidad Católica del Norte, Chile: translated and analyzed ChemCAD models for shellfish cultivation.

2000–2003 · Rowan: archaeological conservation in La Serena, Chile; a statistical program estimating stature from remains.

Leadership & affiliations

President, Temple University SIAM Chapter (2018–2022)

SIAM, AMS and MAA member (2017–2022)

Pi Mu Epsilon (2015–2017)

NJEA and Clearview Regional Education Association (2008–2017)

IEEE member (2006–2008)

Outside work

Reading (neuroscience, mathematics, science fiction, history, philosophy), writing mathematical articles and poetry, drawing and painting, running, weightlifting and hiking, Orioles, Phillies, Ravens and Eagles games, and travel.